WORKED SOLUTION

STEP I 1993, Question 1

Pure • Counting Dice Totals

STEP I 1993, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s1-q1

This question has a full guided treatment in *Beyond STEP*, Section 9.6, *Strange Dice*: https://beyondstep.keithmcnulty.org/ch09/strange-dice.html

The key idea

Every face is a different colour, so a "way" of throwing a total is a choice of one face from each die. If one die has a_k faces numbered k and the other has b_k, the number of ways of throwing a total n is

$$N(n) = \sum_{i+j=n} a_i\, b_j.$$

In the second part we read the table from the smallest totals upwards. Each new total brings in only one or two unknowns we haven't met yet, so the face counts are forced one at a time, and we only have to rule out the few branches that appear along the way.

Solution

The total of 6

On the first die the counts are $a_1 = 1$, $a_2 = 2$, $a_3 = 2$, $a_6 = 1$. On the second they are $b_1 = 1$, $b_3 = 2$, $b_4 = 1$, $b_5 = 1$, $b_6 = 1$. A total of 6 needs the first die to show 1, 2 or 3; a 6 would need a 0 on the second die. So

$$N(6) = a_1b_5 + a_2b_4 + a_3b_3 = 1\cdot 1 + 2\cdot 1 + 2\cdot 2 = 7,$$

which is the seven ways required.

Finding the new numbering

Call the dice P and Q. Let P have a_k faces numbered k, and Q have b_k, for $k = 1, \ldots, 6$. These are non-negative integers with

$$a_1 + \cdots + a_6 = b_1 + \cdots + b_6 = 6.$$

(As a sanity check, the entries in the table add up to $36 = 6 \times 6$.)

Totals 2 and 3. $N(2) = a_1b_1 = 0$, so the two dice can't both have a 1. But $N(3) = a_1b_2 + a_2b_1 = 2$, so some die has a 1. Call that die P. Then $a_1 \geq 1$ and $b_1 = 0$, and

$$N(3) = a_1b_2 = 2, \qquad \text{so} \qquad (a_1, b_2) = (2,1) \text{ or } (1,2).$$

The case $a_1 = 2$, $b_2 = 1$ is impossible. Since Q has no 1, $N(7) \geq a_1b_6 = 2b_6$, and $N(7) = 3$ gives $b_6 \leq 1$. In this case Q has one face numbered 2, and we'll check that it has no 3 or 4:

- $N(4) = a_1b_3 + a_2b_2 = 2b_3 + a_2 = 1$, so $2b_3 \leq 1$, which forces $b_3 = 0$ and then $a_2 = 1$.
- $N(5) = a_1b_4 + a_2b_3 + a_3b_2 = 2b_4 + 0 + a_3 = 1$, so $b_4 = 0$ and $a_3 = 1$.

So Q has one 2, no 1, 3 or 4, and its other five faces are 5s and 6s: $b_5 + b_6 = 5$. Since $b_6 \leq 1$, this gives $b_5 \geq 4$, and then

$$N(6) \geq a_1b_5 = 2b_5 \geq 8,$$

which contradicts $N(6) = 4$. So this case can't happen.

So $a_1 = 1$ and $b_2 = 2$. Now we work up the table, remembering $b_1 = 0$.

- $N(4) = a_1b_3 + a_2b_2 = b_3 + 2a_2 = 1$. Since $2a_2$ is even, $a_2 = 0$ and $b_3 = 1$.
- $N(5) = a_1b_4 + a_2b_3 + a_3b_2 = b_4 + 0 + 2a_3 = 1$. In the same way, $a_3 = 0$ and $b_4 = 1$.
- $N(6) = a_1b_5 + a_2b_4 + a_3b_3 + a_4b_2 = b_5 + 2a_4 = 4$.

So far Q has $b_1 + b_2 + b_3 + b_4 = 0 + 2 + 1 + 1 = 4$ faces accounted for, and so $b_5 + b_6 = 2$. From $b_5 = 4 - 2a_4$, b_5 is even, so $b_5 = 0$ or $b_5 = 2$.

? **Ask yourself** How do we decide between $b_5 = 0$ and $b_5 = 2$?

We use the next total. P has no 2 or 3 and Q has no 1, so

$$N(7) = a_1b_6 + a_4b_3 + a_5b_2 = b_6 + a_4 + 2a_5 = 3.$$

- If $b_5 = 0$, then $a_4 = 2$ and $b_6 = 2$. Now $N(7) \geq b_6 + a_4 = 4 > 3$, which is impossible.
- If $b_5 = 2$, then $a_4 = 1$ and $b_6 = 0$. Now $N(7) = 0 + 1 + 2a_5 = 3$ gives $a_5 = 1$. P has $a_1 + a_4 + a_5 = 3$ faces accounted for, so $a_6 = 3$.

The table has forced everything:

$$P: \ 1, 4, 5, 6, 6, 6 \qquad\qquad Q: \ 2, 2, 3, 4, 5, 5.$$

Checking the rest of the table. We used only the totals 2 to 7, so we still have to confirm that this numbering gives the remaining entries. With $a_1 = a_4 = a_5 = 1$, $a_6 = 3$ and $b_2 = 2$, $b_3 = b_4 = 1$, $b_5 = 2$:

$$\begin{aligned} N(8) &= a_4b_4 + a_5b_3 + a_6b_2 = 1 + 1 + 6 = 8, \\ N(9) &= a_4b_5 + a_5b_4 + a_6b_3 = 2 + 1 + 3 = 6, \\ N(10) &= a_5b_5 + a_6b_4 = 2 + 3 = 5, \\ N(11) &= a_6b_5 = 6, \\ N(12) &= a_6b_6 = 0. \end{aligned}$$

All of these agree with the table. So the dice must be numbered $1,4,5,6,6,6$ and $2,2,3,4,5,5$, and this numbering really does give the table. It is the only possible numbering, apart from swapping which die we call which.

Checks

- Draw the 6×6 grid of sums for $1,4,5,6,6,6$ against $2,2,3,4,5,5$ and count each total. You should get $0,2,1,1,4,3,8,6,5,6,0$.
- A computer search over every way of numbering two dice with 1 to 6 (there are 462 multisets per die) finds exactly this pair and nothing else, which confirms that the solution is unique.

Common slips

- Guessing a numbering that fits the table and stopping. The question asks for reasoning, so you need to show that the table forces the numbering.
- Forgetting the case $a_1 = 2$, $b_2 = 1$, or dismissing it without a reason.
- Fixing the face counts from the small totals and never checking the totals 8 to 12.

Beyond STEP

The neat way to package $N(n)$ is with a generating function. If $p(x) = \sum a_k x^k$ and $q(x) = \sum b_k x^k$, then the coefficient of x^n in $p(x)q(x)$ is exactly $N(n)$, so the question asks you to factorise a given polynomial into two pieces of a particular shape. The same idea gives the *Sicherman dice*, $1,2,2,3,3,4$ and $1,3,4,5,6,8$, which have the same distribution of totals as two ordinary dice. Herbert Wilf's *generatingfunctionology* is a lively introduction to the method.

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